

Kavya Nambiar

NLP Researcher (PhD)

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Profile

NLP researcher completing a PhD at IISc Bangalore on efficient transformer inference (defence completed June 2026). Published at ACL and EMNLP on KV-cache compression and speculative decoding, with additional work on Indic-language evaluation. Looking to move from research into applied LLM engineering.

Skills

Research: Transformer architectures, efficient inference (quantization, KV-cache compression, speculative decoding), retrieval-augmented generation, Indic-language evaluation methodology

Stack: Python, PyTorch, Hugging Face, CUDA (basic), vLLM (research use), LaTeX, Git

Experience

PhD Researcher, NLP — Indian Institute of Science (CSA), Bengaluru

Aug 2022 – Jun 2026

- Thesis: "Memory-Efficient Inference for Autoregressive Language Models" — three first-author papers (ACL 2025, EMNLP 2024, ACL 2024 Findings).
- Proposed a KV-cache compression method achieving 3.1x memory reduction at <0.5% quality loss; adopted by two open-source serving projects.
- Teaching assistant for the M.Tech "Natural Language Understanding" course (3 years).

Research Intern — Microsoft Research India, Bengaluru

Jun 2025 – Sep 2025

- Worked on inference-efficiency benchmarks for production serving stacks.

Research Intern — VaakAI (Indic voice-LLM startup), Bengaluru

Jun 2024 – Sep 2024

- Studied retrieval quality effects on RAG hallucination rates for Hindi/Kannada queries (workshop paper).

Education

PhD, Computer Science (NLP) — Indian Institute of Science, Bengaluru, 2022 – 2026

M.Tech Computational Science — Indian Institute of Science, Bengaluru, 2020 – 2022

B.Tech Computer Science & Engineering — NIT Calicut, 2016 – 2020

Selected publications

- "Sublinear KV-Cache Growth for Long-Context Inference" — ACL 2025 (main).
- "When Does Speculative Decoding Help? A Systematic Study" — EMNLP 2024.
- "Retrieval Noise and Hallucination in Indic-Language RAG" — RepL4NLP workshop 2024.

Languages known

Malayalam (native), English (fluent), Hindi (working), Kannada (working)